

UNALAKLEET, AK, USA

WMO: 702070

Lat: 63.883N

Lon: 160.800W

Elev: 6

StdP: 101.26

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26627

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-34.0	-30.9	-34.0	0.2	-28.8	-33.0	0.2	-27.2	19.1	-11.0	17.5	-10.6	1.7	130	0.869

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.5	21.0	14.6	18.7	14.0	17.3	12.9	15.6	19.2	14.6	18.0	13.5	16.7	4.9	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.8	9.8	17.4	12.6	9.1	16.3	12.0	8.7	15.6	43.7	19.4	40.6	17.9	37.8	16.6	20.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.3	13.5	11.9	DB	-36.0	25.7	4.9	2.4	-39.5	27.5	-42.4	28.9	-45.1	30.3	-48.7	32.0	(4)
(5)			WB	-33.8	16.8	3.0	1.7	-35.9	18.1	-37.7	19.0	-39.4	20.0	-41.6	21.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.3	-13.8	-10.6	-11.3	-3.2	4.1	10.0	13.1	11.4	7.4	-1.4	-10.0	-12.0		
	DBStd	11.59	9.77	7.96	7.70	6.52	4.71	3.58	2.58	2.94	3.49	5.78	7.22	7.93		
	HDD10.0	4348	737	576	660	397	187	42	4	18	89	354	601	683		
	HDD18.3	7173	996	809	918	647	441	250	164	216	328	613	851	941		
	CDD10.0	218	0	0	0	0	5	42	99	61	11	0	0	0		
	CDD18.3	2	0	0	0	0	0	1	1	1	0	0	0	0		
	CDH23.3	18	0	0	0	0	1	9	7	2	0	0	0	0		
	CDH26.7	2	0	0	0	0	0	2	1	0	0	0	0	0		
(14) Wind	WSAvg	5.1	6.6	6.9	5.2	4.7	3.8	3.7	4.2	4.4	4.5	4.9	6.0	6.4		
(15) Precipitation	PrecAvg	327	12	9	12	15	18	34	57	73	57	26	17	18		
	PrecMax	501	50	30	56	51	64	126	113	189	167	73	51	50		
	PrecMin	203	1	0	0	0	0	6	0	28	9	5	0	1		
	PrecStd	91	11	9	13	14	16	28	32	37	44	16	13	13		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.0	3.9	4.9	11.3	19.0	23.9	24.0	22.3	17.6	12.0	3.9	3.6		
		MCWB	1.8	1.3	1.7	5.2	10.4	13.7	15.5	15.7	11.4	7.8	1.7	1.4		
	2%	DB	2.5	2.5	2.7	8.7	14.7	19.4	21.1	19.1	14.9	8.5	2.5	2.1		
		MCWB	0.4	0.5	0.2	4.1	8.6	13.5	15.2	14.4	10.2	6.0	0.7	0.4		
	5%	DB	0.9	1.3	1.2	6.5	12.4	17.3	18.8	17.6	13.0	7.3	1.3	0.0		
		MCWB	-1.2	-0.5	-0.9	2.7	7.8	12.2	14.8	13.4	9.8	4.9	-0.3	-1.5		
	10%	DB	-1.4	-0.9	-1.0	4.4	10.9	15.9	17.4	16.2	12.3	6.0	-0.8	-2.2		
		MCWB	-2.7	-2.2	-2.5	1.5	6.8	11.8	13.4	12.7	9.4	3.8	-1.9	-3.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.0	1.9	2.0	5.7	11.6	16.0	17.8	16.7	12.3	7.9	2.5	1.9		
		MCDB	3.9	3.7	4.5	10.1	17.8	21.1	20.6	20.5	15.3	10.4	3.6	3.1		
	2%	WB	0.7	0.7	0.6	4.2	9.7	14.0	16.0	15.2	11.0	6.4	0.9	0.3		
		MCDB	2.4	2.3	2.5	8.1	13.7	18.0	19.4	18.2	13.8	8.2	2.4	1.8		
	5%	WB	-1.1	-0.5	-0.8	3.0	8.1	12.9	15.0	14.1	10.3	5.1	-0.3	-1.5		
		MCDB	0.3	1.0	1.0	6.4	11.8	16.8	18.5	17.1	12.9	7.1	1.0	-0.2		
	10%	WB	-3.0	-2.0	-2.4	1.8	6.9	11.8	14.0	13.0	9.6	3.9	-1.8	-3.4		
		MCDB	-1.7	-0.7	-1.0	4.5	10.0	15.4	17.2	15.5	12.0	6.0	-0.6	-2.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.8	6.2	7.5	7.3	6.8	6.3	5.5	6.0	6.0	5.5	5.9	5.7		
		MCDBR	5.5	5.7	6.5	8.1	10.7	10.4	9.4	8.6	7.6	5.3	5.1	5.1		
	5% WB	MCWBR	4.8	4.9	5.3	5.5	6.4	5.9	5.4	5.3	4.9	4.0	4.4	4.3		
		MCWBR	5.8	5.8	6.1	7.8	9.7	9.1	8.0	7.5	6.5	4.7	5.0	4.9		
(40) Clear-Sky Solar Irradiance	taub		0.194	0.262	0.275	0.337	0.365	0.370	0.366	0.385	0.319	0.284	0.196	0.188		
		taud	1.951	2.155	2.221	2.110	2.255	2.358	2.416	2.360	2.527	2.370	1.964	1.897		
	Ebn at Noon		397	661	822	829	842	847	840	781	782	642	374	159		
		Edh at Noon	28	60	86	122	116	108	99	96	66	49	26	13		
(44) All-Sky Solar Radiation	RadAvg		0.19	0.82	2.58	4.38	5.03	5.32	4.41	3.40	2.15	0.94	0.27	0.07		
	RadStd		0.03	0.11	0.25	0.28	0.45	0.52	0.30	0.42	0.22	0.13	0.04	0.01		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (6 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page